

ICC-Tutorial: Worksheet for Abstraction Using Interval Neural Networks (INNs)

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1 Worksheet: Abstraction Using Interval Neural Networks (INNs)

Goal

- Replace a concrete NN with a smaller, abstract structure.
- Same structure as an NN, but have intervals for weights and biases.
- Obtained by merging nodes of the given NN
- Concrete NN: concrete neurons/weights/biases; INN: abstract neurons/interval weights/interval biases.

Why INNs?

- Large NNs are expensive to analyze directly.
- Abstraction reduces size while over-approximating behavior.
- Group neurons $\Rightarrow$ simplified architecture.
- Intervals $\Rightarrow$ behavioral over-approximation.

Abstraction Procedure (With Example)

Consider the NN in Fig. 1 (2 inputs, 4+4 hidden, 2 outputs).

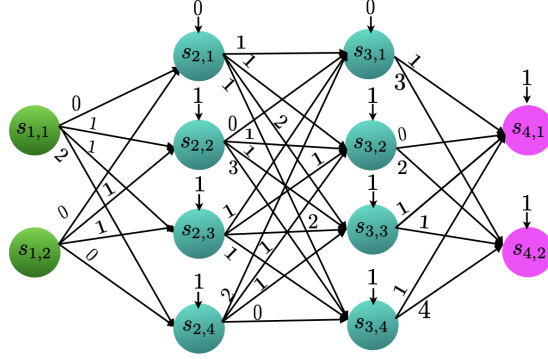


Figure 1: Neural Network $\mathcal{N}$

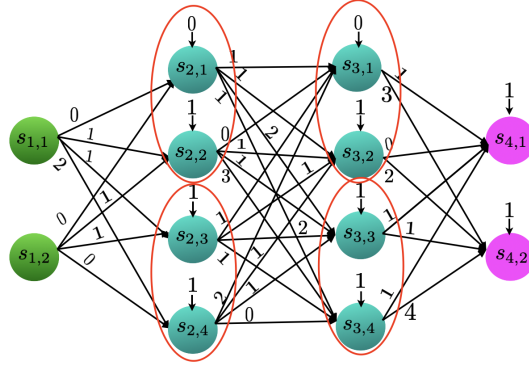


Figure 2: Neural Network $\mathcal{N}$

1. Choose a Partition

- Input and output layers: not partitioned.
- Hidden layers grouped in pairs:

$$P_1 = \{\{s_{2,1}, s_{2,2}\}, \{s_{2,3}, s_{2,4}\}\}, \quad P_2 = \{\{s_{3,1}, s_{3,2}\}, \{s_{3,3}, s_{3,4}\}\}.$$

2. Assign Interval Biases

- For a block with biases $b_1, \dots, b_m$:

$$\text{bias} = [\min\{b_i\}, \max\{b_i\}].$$

- Example: block $\{s_{2,1}, s_{2,2}\}$ has biases 0 and 1 $\Rightarrow$ abstract bias = $[0, 1]$.

3. Assign Interval Weights

- To define the interval weight from one block to another block, we take all the weights connecting nodes in the source block to nodes in the target block, compute their minimum and maximum values, and then multiply these bounds by the number of nodes in the source block.
- Example: Consider the weights from the block $\{s_{2,1}, s_{2,2}\}$ to the block $\{s_{3,1}, s_{3,2}\}$:

$$w_{s_{2,1}, s_{3,1}} = 1, \quad w_{s_{2,1}, s_{3,2}} = 1, \quad w_{s_{2,2}, s_{3,1}} = 0, \quad w_{s_{2,2}, s_{3,2}} = 1.$$

The minimum and maximum of these weights are 0 and 1. Since the source block contains 2 nodes, we multiply the bounds by 2. Hence, the interval weight from the first block to the second block is $[0, 2]$.

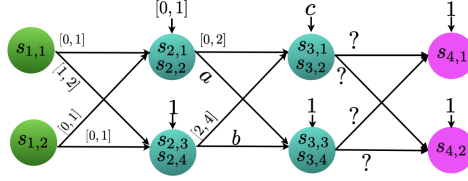


Figure 3: Abstraction of $\mathcal{N}$

2 Exercises

Exercise 1

- Find values of a , b and c .
- Complete the INN shown in Figure 3, i.e., all the weights and biases.

Exercise 2: Constructing a different INN We now change the partition of the hidden layers and construct a new INN. For the same NN shown in Fig. 1, consider the following **different** grouping of neurons.

Hidden Layer 1:

$$P'_1 = \{\{s_{2,1}, s_{2,4}\}, \{s_{2,2}, s_{2,3}\}\}.$$

Hidden Layer 2:

$$P'_2 = \{\{s_{3,1}, s_{3,3}\}, \{s_{3,2}, s_{3,4}\}\}.$$

Task 1: Construct the abstract nodes.

- Compute the interval bias of $\{s_{2,1}, s_{2,3}\}$

Task 2: Compute interval weights.

Task 3: Draw your INN.

Notes on Semantics of an AINN

- Represents a family of NNs.
- Each weight/bias chosen from its interval.
- Executing an AINN = executing any NN consistent with these intervals.
- One input $\Rightarrow$ a set of possible valuations (not a single point).

Notes

- Many abstractions possible: depends on partitions.
- Coarse partition: smaller abstraction, more conservative.
- Fine partition: larger abstraction, more precise.
- INN needs:
 1. Original NN,
 2. A chosen partition for each layer.

Example: Forward Propagation in an AINN

Consider the AINN in Fig. 3. Input to the network:

$$x = (1, 1),$$

so value at $s_{0,1} = 1$, and value at $s_{0,2} = 1$.

First Hidden Layer

Node $\{s_{2,1}, s_{2,2}\}$ Contribution from $s_{1,1} = 1 \times [0, 1] = [0, 1]$, and contribution from $s_{1,2} = 1 \times [0, 1] = [0, 1]$

Bias interval: $[0, 1]$.

Total:

$$[0, 1] + [0, 1] + [0, 1] = [0, 3].$$

ReLU:

$$\text{ReLU}([0, 3]) = [0, 3].$$

Node $\{s_{2,3}, s_{2,4}\}$ Weights: $[1, 2]$ from $s_{1,1}$, and $[0, 1]$ from $s_{1,2}$.

$$1 \times [1, 2] \times 1 = [1, 2], \quad 1 \times [0, 1] \times 1 = [0, 1]$$

Include bias $[1, 1]$:

$$[1, 2] + [0, 1] + [1, 1] = [2, 4].$$

ReLU:

$$\text{ReLU}([2, 4]) = [2, 4].$$

Hence $V_1 = [0, 3] \times [2, 4]$.

Second Hidden Layer

Pick a single valuation from V_1 :

$$y_1 = (1, 2),$$

i.e. value 1 at $\{s_{2,1}, s_{2,2}\}$ and 2 at $\{s_{2,3}, s_{2,4}\}$.

Node $\{s_{3,1}, s_{3,2}\}$ Incoming Weights: $[0, 2]$ and $[2, 4]$

Compute:

$$1 \times [0, 2] = [0, 2], \quad 2 \times [2, 4] = [4, 8].$$

Add bias $[0, 1]$:

$$[0, 2] + [4, 8] + [0, 1] = [4, 11].$$

ReLU does not change.

Conclusion: To obtain the full valuation set for layer 2, propagate all values in

$$V_1 = [0, 3] \times [2, 4].$$

Exercise.

- Using the same chosen valuation $y_1 = (1, 2)$, compute the interval value at the second abstract neuron of layer 2, $\{s_{3,3}, s_{3,4}\}$.

Write your final answer as:

$$y = [\text{_____}, \text{_____}].$$

Soundness of the Abstraction

- Abstraction preserves all behaviors of the concrete NN.
- If (x, u) is achievable by the original NN, it is also achievable by the INN.
- INN over-approximates: it may allow more behaviors but never loses any real one.

Computing Input–Output Sets of an AINN

- AINN = family of many NNs $\Rightarrow$ compute all possible outputs.
- Tools for this:
 - MILP-based analysis
 - Star-set symbolic propagation
- Both compute reachable sets layer-by-layer using interval semantics.